

Wireless Protocols for IoT Part I: Bluetooth and Bluetooth Smart



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Audio/Video recordings of this class lecture are available on-line at:

<http://www.cse.wustl.edu/~jain/cse574-16/>

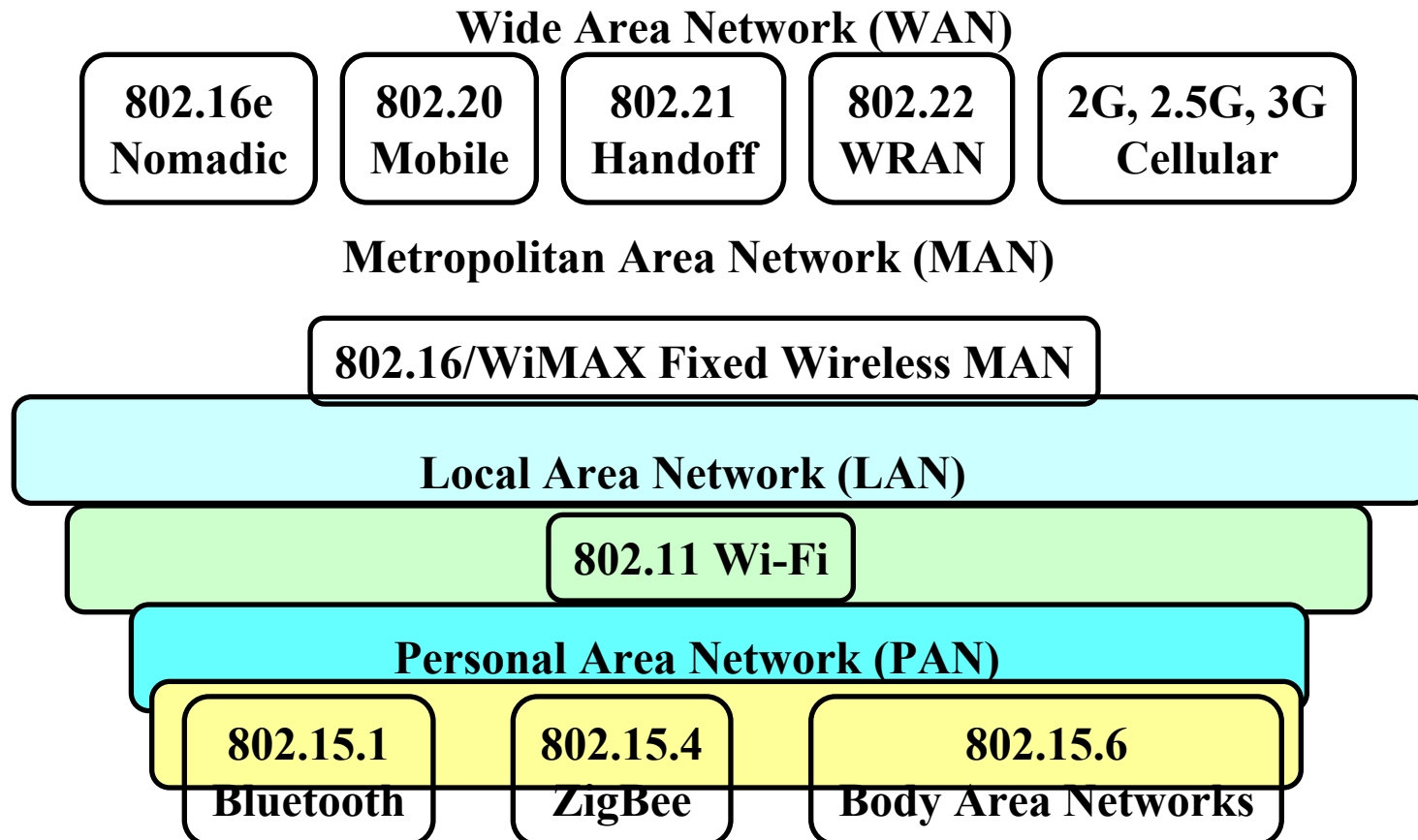


1. Wireless Personal Area Networks (WPANs)
2. IEEE 802.15 Projects
3. Bluetooth: Packet Format, Energy Management
4. Bluetooth Protocol Stack, Application Profiles
5. Bluetooth LE: Protocol Stack, PHY, MAC
6. Bluetooth and WiFi Coexistence

Note: This is 1st in a series of lectures on WPANs. ZigBee and other networks are discussed in subsequent lectures.

Wireless Personal Area Networks (WPANs)

- 10m or less



WPAN: Design Challenges

- ❑ **Battery powered:** Maximize battery life.
A few hours to a few years on a coin cell.
- ❑ **Dynamic topologies:** Short duration connections and then device is turned off or goes to sleep
- ❑ **No infrastructure**
- ❑ **Avoid Interference** due to larger powered LAN devices
- ❑ **Simple and Extreme Interoperability:** Billions of devices.
More variety than LAN or MAN
- ❑ **Low-cost:** A few dollars



IEEE 802.15 Projects

- ❑ **IEEE 802.15.1-2005: Bluetooth 1.2**
- ❑ **IEEE 802.15.2-2003**: Coexistence Recommended Practice
- ❑ **IEEE 802.15.3-2003**: High Rate (55 Mbps) Multimedia WPAN
- ❑ IEEE 802.15.3a: Ultra-Wide Band Phy - disbanded
- ❑ **IEEE 802.15.3b-2005**: MAC Interoperability
- ❑ **IEEE 802.15.3c-2009**: High Rate (>1Gbps) mm Wave PHY
- ❑ IEEE 802.15.3d: 100 Gbps point-to-point Phy
- ❑ IEEE 802.15.3e: High-Rate close proximity point-to-point MAC and PHY
- ❑ **IEEE 802.15.4-2011**: Low Rate (250kbps) WPAN - **ZigBee**
- ❑ **IEEE 802.15.4e-2012**: MAC Enhancements

IEEE 802.15 Projects (Cont)

- ❑ IEEE 802.15.4f-2012: PHY for Active RFID
- ❑ IEEE 802.15.4g-2012: PHY for Smart Utility Networks
- ❑ IEEE 802.15.4h: 802.15.4 Corrigendum 1
- ❑ IEEE 802.15.4i: 802.15.4 Roll-up to include 15.4a, c & d
- ❑ IEEE 802.15.4j-2013: Medical Body Area Network 2.36-2.4 GHz
- ❑ IEEE 802.15.4k-2013: Low Energy Critical Infrastructure Monitoring PHY
- ❑ IEEE 802.15.4m-2014: TV White Spaces PHY
- ❑ IEEE P802.15.4n: China Medical Band PHY
- ❑ IEEE P802.15.4p-2014: Positive Train Control (Rail Communications & Control) PHY
- ❑ IEEE P802.15.4q: Ultra Low Power PHY

IEEE 802.15 Projects (Cont)

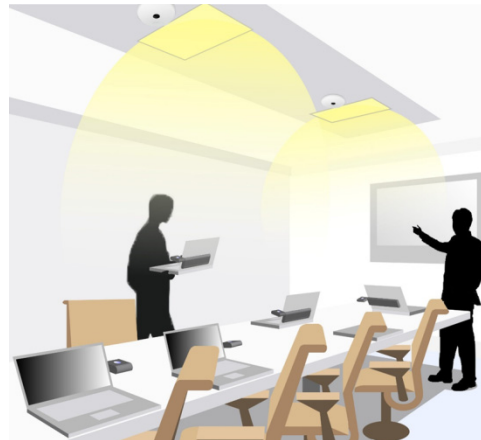
- ❑ IEEE 802.15.4r: Common Ranging (Distance Measurement) Protocol
- ❑ IEEE 802.15.4s: System resource usage
- ❑ IEEE 802.15.4t: High rate PHY
- ❑ IEEE 802.15.4u: 865-867 MHz band in India
- ❑ IEEE 802.15.5-2009: Mesh Networking. Full/partial meshes. Range Extension
- ❑ IEEE 802.15.6-2012: Body Area Networking. Medical and entertainment. Low power
- ❑ IEEE 802.15.7-2011: Visible Light Communications
- ❑ IEEE 802.15.7r1: Optical wireless (infrared, ultraviolet, visible light)
- ❑ IEEE P802.15.8: Peer Aware Communications

IEEE 802.15 Projects (Cont)

- ❑ IEEE P802.15.9: Key Management Protocol
- ❑ IEEE P802.15.10: Layer 2 (Mesh) Routing
- ❑ IEEE 802.15 SG12: Consolidated Link Layer Control study group
- ❑ IEEE 802.15 IG6T: Consolidate Link Layer Control interest group
- ❑ IEEE 802.15 IG_{dep}: Enhanced Dependability interest group
- ❑ IEEE 802.15 IG_{guide}: Guide for 15.4 use interest group
- ❑ IEEE 802.15 IG_{hrrc}: High Rate Rail Communications interest group
- ❑ IEEE 802.15 IT_{THz}: Terahertz interest group
- ❑ IEEE 802.15 SC_{wng}: Wireless Next-Generation standing committee
- ❑ IEEE 802.15 SC_{maint}: Maintenance standing committee

Li-Fi

- ❑ Optical wireless using light
- ❑ Localized $\Rightarrow$ Secure
- ❑ High SNR $\Rightarrow$ Dense reuse
- ❑ No electromagnetic interference $\Rightarrow$ Good for EMI sensitive environments, e.g., planes



Ref: N. Serafimovski, "What is Li-Fi?," Jan 2015, <https://mentor.ieee.org/802.15/dcn/15/15-15-0107-01-007a-lifi-definition.pptx>
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Bluetooth

- ❑ Started with Ericsson's Bluetooth Project in 1994 for radio-communication between cell phones over short distances
- ❑ Named after Danish king Harald Blatand (AD 940-981) who was fond of blueberries
- ❑ Intel, IBM, Nokia, Toshiba, and Ericsson formed Bluetooth SIG in May 1998
- ❑ Version 1.0A of the specification came out in late 1999.
- ❑ IEEE 802.15.1 approved in early 2002 is based on Bluetooth
Later versions handled by Bluetooth SIG directly
- ❑ Key Features:
 - Lower Power: 10 mA in standby, 50 mA while transmitting
 - Cheap: \$5 per device
 - Small: 9 mm² single chips



Bluetooth Versions

- ❑ **Bluetooth 1.1**: IEEE 802.15.1-2002
- ❑ **Bluetooth 1.2**: IEEE 802.15.1-2005. Completed Nov 2003. Extended SCO, Higher variable rate retransmission for SCO + Adaptive frequency hopping (avoid frequencies with interference).
- ❑ **Bluetooth 2.0** + Enhanced Data Rate (EDR) (Nov 2004): 3 Mbps using DPSK. For video applications. Reduced power due to reduced duty cycle
- ❑ **Bluetooth 2.1** + EDR (July 2007): Secure Simple Pairing to speed up pairing
- ❑ **Bluetooth 3.0** + High Speed (HS) (April 2009): 24 Mbps using WiFi PHY + Bluetooth PHY for lower rates
- ❑ **Bluetooth 4.0** (June 2010): Low energy. Smaller devices requiring longer battery life (several years). New incompatible PHY. Bluetooth Smart or BLE
- ❑ **Bluetooth 4.1**: 4.0 + Core Specification Amendments (CSA) 1, 2, 3, 4
- ❑ **Bluetooth 4.2** (Dec 2014): Larger packets, security/privacy, IPv6 profile

Ref: ITL, "Security of Bluetooth Systems and Devices," http://csrc.nist.gov/publications/nistbul/august-2012_itl-bulletin.pdf

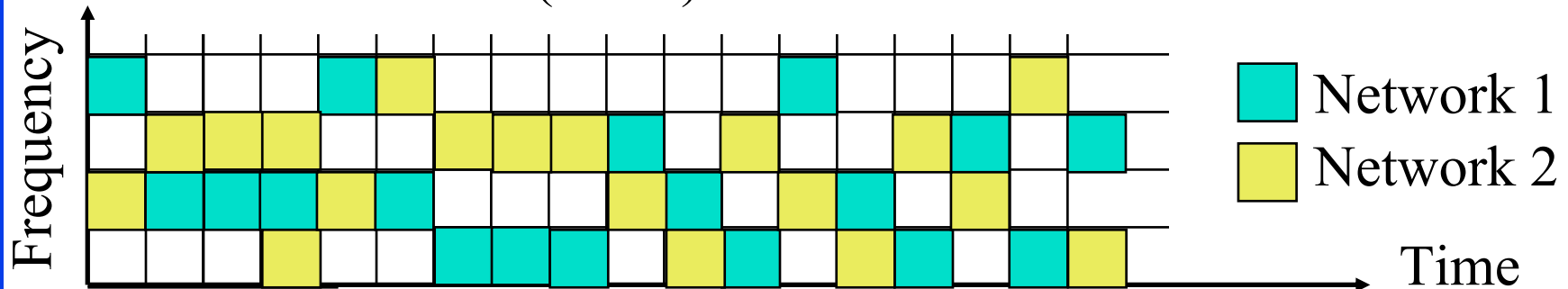
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Bluetooth: Details

- ❑ **Frequency Range:** 2402 - 2480 MHz
(total 79 MHz band) 23 MHz in some countries, e.g., Spain
- ❑ **Data Rate:** 1 Mbps using 1 MHz (Nominal) 720 kbps (User)
- ❑ **Radio Frequency hopping:** 1600 times/s $\Rightarrow$ 625 ms/hop
- ❑ **Security:** Challenge/Response Authentication. 128b Encryption
- ❑ **TX Output Power:**
 - Class 1: 20 dBm Max. (0.1 W) – 100m
 - Class 2: 4 dBm (2.5 mW)
 - **Class 3:** 0 dBm (1mW) – 10m



Ref: <http://www.bluetooth.com/>, <http://www.bluetooth.org/>, <http://grouper.ieee.org/groups/802/15/index.html>

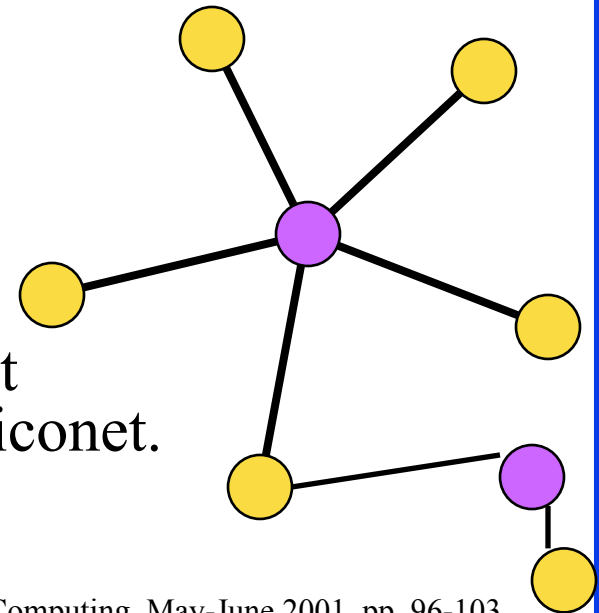
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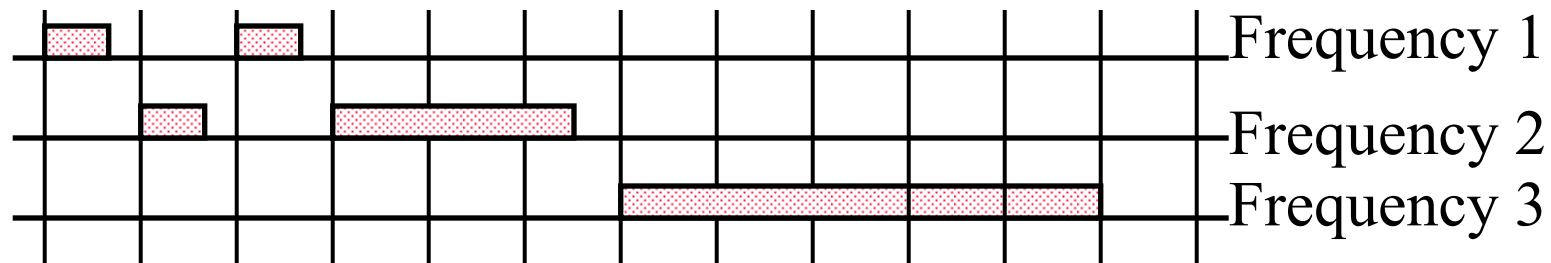
Piconet

- ❑ Piconet is formed by a master and many slaves
 - Up to 7 active slaves.
Slaves can only transmit when requested by master
 - Up to 255 Parked slaves
- ❑ Active slaves are polled by master for transmission
- ❑ Each station gets a 8-bit parked address
⇒ 255 parked slaves/piconet
- ❑ The parked station can join in 2ms.
- ❑ Other stations can join in more time.
- ❑ **Scatter net**: A device can participate in multiple Pico nets ⇒ Timeshare and must synchronize to the master of the current piconet.
Routing protocol not defined.



Ref: P. Bhagwat, "Bluetooth Technology for short range wireless Apps," IEEE Internet Computing, May-June 2001, pp. 96-103, [bluetooth.pdf \(Must read\)](#)

Frequency Hopping Sequences



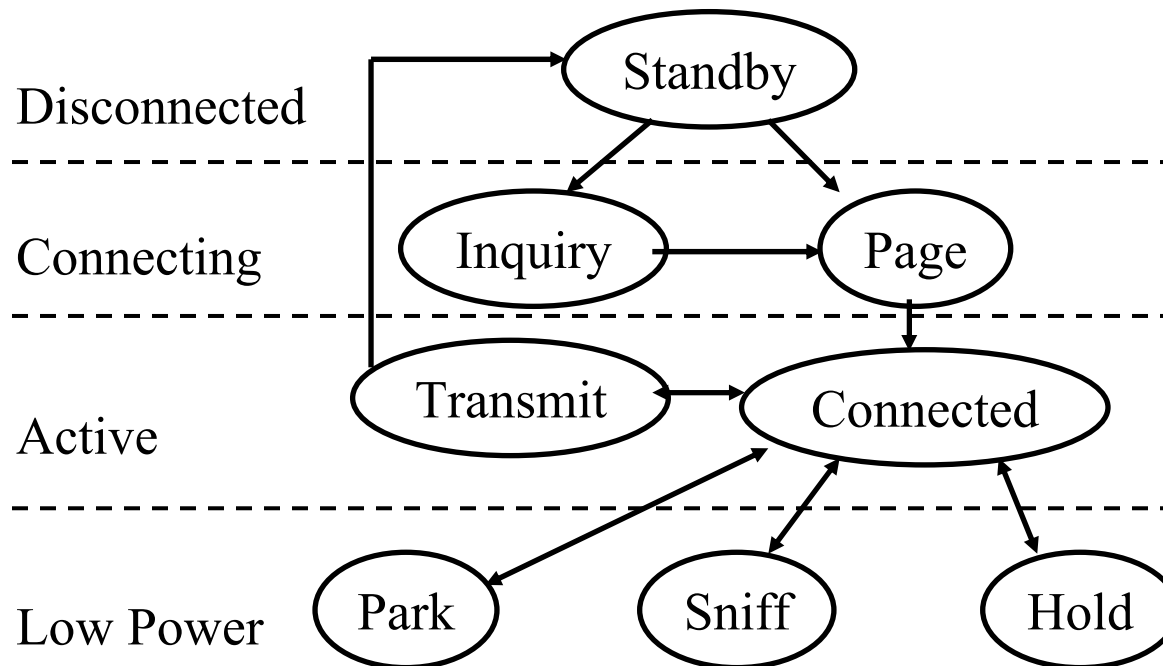
- ❑ 625 ms slots using a 312.5 ms clock
- ❑ Time-division duplex (TDD)
⇒ Downstream and upstream alternate
- ❑ Master starts in even numbered slots only.
- ❑ Slaves start in odd numbered slots only
- ❑ Slaves can transmit in one slot right after receiving a packet from master
- ❑ Packets = 1 slot, 3 slot, or 5 slots long
- ❑ The frequency hop is skipped during a packet.

Bluetooth Packet Format

Access Code	Baseband/Link Control Header	Data Payload
72b	54b	0-2745b

- ❑ Packets can be up to five slots long. 5 slots = 3125 bits.
- ❑ Access codes:
 - Channel access code identifies the piconet
 - Device access code for paging requests and response
 - Inquiry access code to discover units
- ❑ Header: member address (3b), type code (4b), flow control, ack/nack (1b), sequence number, and header error check (8b)
18b Header is encoded using 1/3 rate FEC resulting in 54b
- ❑ Synchronous traffic has periodic reserved slots.
- ❑ Other slots can be allocated for asynchronous traffic

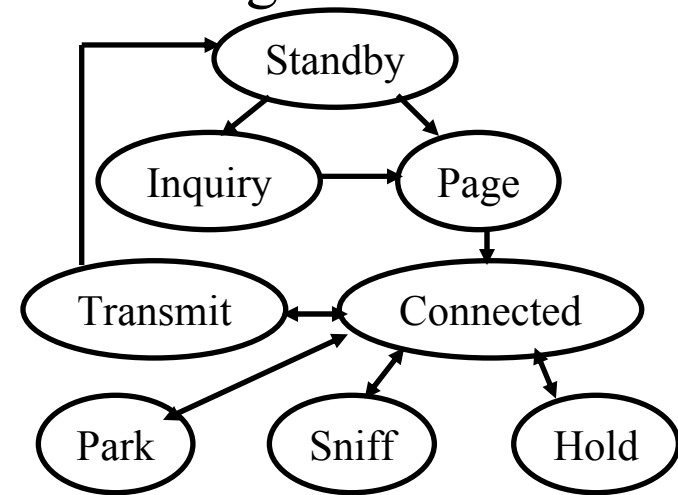
Bluetooth Operational States



- ❑ **Standby:** Initial state
- ❑ **Inquiry:** Master sends an inquiry packet. Slaves scan for inquiries and respond with their address and clock after a random delay (CSMA/CA)

Bluetooth Operational States (Cont)

- ❑ **Page:** Master in page state invites devices to join the piconet. Page message is sent in 3 consecutive slots (3 frequencies). Slave enters page response state and sends page response including its device access code.
- ❑ Master informs slave about its clock and address so that slave can participate in piconet. Slave computes the clock offset.
- ❑ **Connected:** A short 3-bit logical address is assigned
- ❑ **Transmit:**

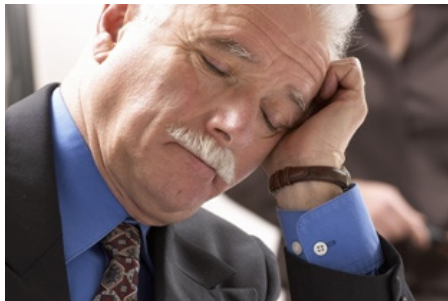


Energy Management in Bluetooth

Three inactive states:

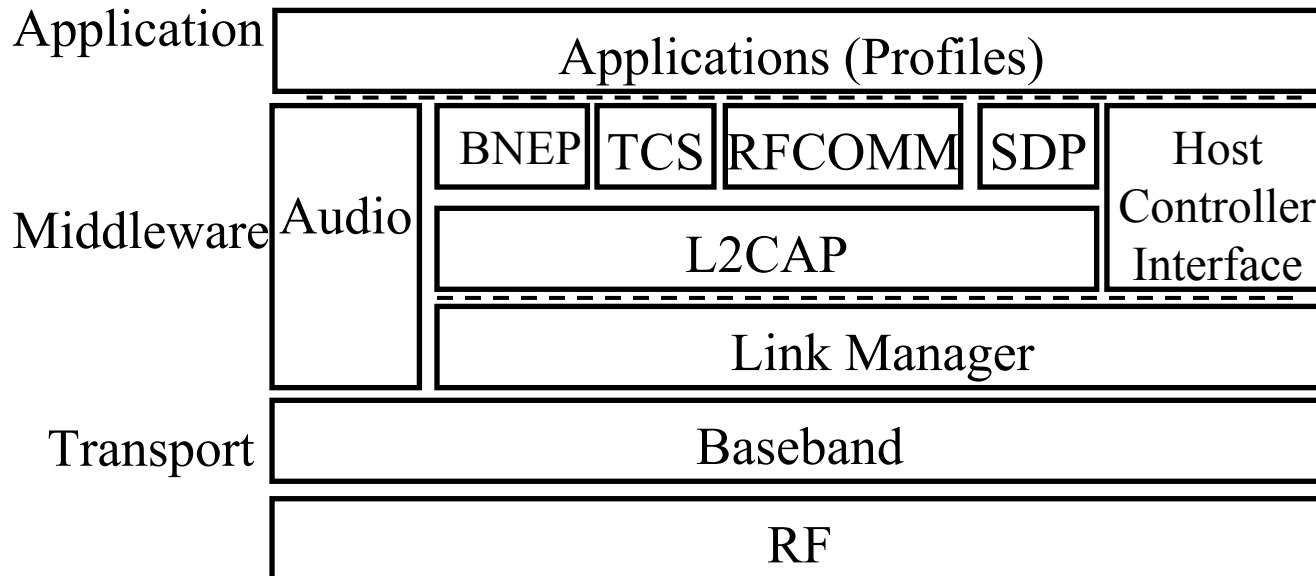
1. **Hold**: No Asynchronous Connection List (ACL). Synchronous Connection Oriented (SCO) continues.
Node can do something else: scan, page, inquire
2. **Sniff**: Low-power mode. Slave listens after fixed sniff intervals.
3. **Park**: Very Low-power mode. Gives up its 3-bit active member address and gets an 8-bit parked member address. Wake up periodically and listen to beacons. Master broadcasts a train of beacons periodically

Sniff

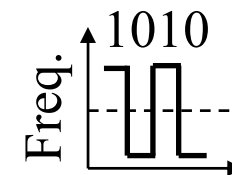


Park

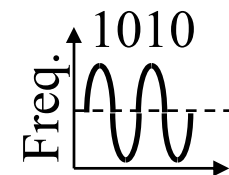
Bluetooth Protocol Stack



- ❑ **RF:** Frequency hopping Gaussian Frequency Shift Keying (GFSK) modulation
- ❑ **Baseband:** Frequency hop selection, connection, MAC



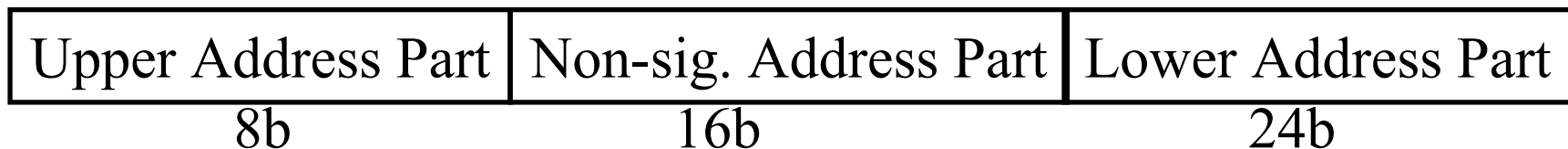
Time
FSK



Time
GFSK

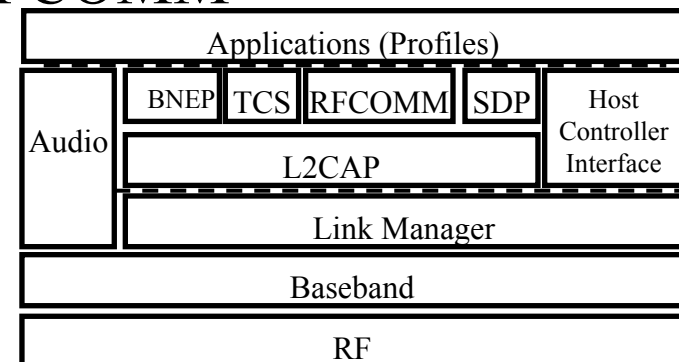
Baseband Layer

- ❑ Each device has a 48-bit IEEE MAC address
- ❑ 3 parts:
 - Lower address part (LAP) – 24 bits
 - Upper address part (UAP) – 8 bits
 - Non-significant address part (NAP) - 16 bits
- ❑ UAP+NAP = Organizationally Unique Identifier (OUI) from IEEE
- ❑ LAP is used in identifying the piconet and other operations
- ❑ Clock runs at 3200 cycles/sec or 312.5 ms (twice the hop rate)



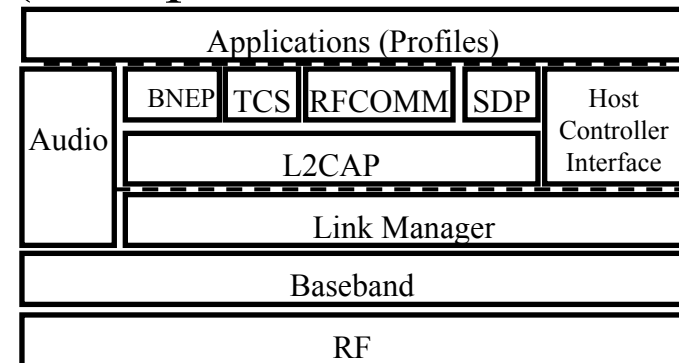
Bluetooth Protocol Stack (Cont)

- ❑ **Link Manager:** Negotiate parameters, Set up connections
- ❑ **Logical Link Control and Adaptation Protocol (L2CAP):**
 - Protocol multiplexing
 - Segmentation and reassembly
 - Controls peak bandwidth, latency, and delay variation
- ❑ **Host Controller Interface:** Chip independent interface to Bluetooth chip. Allows same software to run on all chips.
- ❑ **RFCOMM Layer:** Presents a virtual serial port
 - Sets up a connection to another RFCOMM
- ❑ **Service Discovery Protocol (SDP):**
Devices can discover the services offered and their parameters



Bluetooth Protocol Stack (Cont)

- ❑ **Bluetooth Network Encapsulation Protocol (BNEP):** To transport Ethernet/IP packets over Bluetooth
- ❑ **IrDA Interoperability protocols:** Allow existing IrDA applications to work w/o changes. IrDA object Exchange (IrOBEX) and Infrared Mobile Communication (IrMC) for synchronization
- ❑ **Audio** is carried over 64 kbps over SCO links over baseband
- ❑ **Telephony control specification binary (TCS-BIN):** Call control including group management (multiple extensions, call forwarding, and group calls)
- ❑ **Application Profiles:** Set of algorithms, options, and parameters.



Application Profile Examples

- ☐ Headset Profile
- ☐ Global Navigation Satellite System Profile
- ☐ Hands-Free Profile
- ☐ Phone Book Access Profile
- ☐ SIM Access Profile
- ☐ Synchronization Profile
- ☐ Video Distribution Profile
- ☐ Blood Pressure Profile
- ☐ Cycling Power Profile
- ☐ Find Me Profile
- ☐ Heart Rate Profile
- ☐ Basic Printing Profile
- ☐ Dial-Up Networking Profile
- ☐ File Transfer Profile

Ref: Bluetooth SIGn, “Adopted Bluetooth Profiles, Services, Protocols and Transports,”

<https://www.bluetooth.org/en-us/specification/adopted-specifications>

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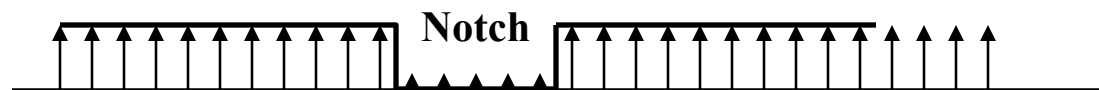
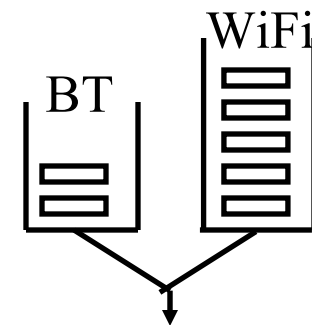
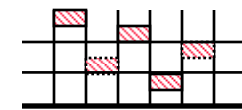
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Bluetooth and WiFi Coexistence

- ❑ Bluetooth frequency hops in 1 MHz carriers over 2402 - 2480 MHz (79 MHz total)
- ❑ WiFi uses OFDM with 52 subcarriers in 20 MHz channels in 2402-2480 MHz (3 non-overlapping channels)
- ❑ Most computers have both Bluetooth and WiFi
- ❑ Collaborative Strategies: Two networks on the same device
- ❑ Non-Collaborative Strategies: No common device

Collaborative Coexistence Strategies

- ❑ Both networks on the same equipment (Laptop or iPhone):
 1. Time Division: Bluetooth skips slots when WiFi is busy, WiFi reserves time for Bluetooth between Beacons
 2. Packet Traffic Arbitration: Packets are prioritized and queued on a common queue for transmission
 3. Notch Filter: WiFi OFDM does not use subcarriers to which Bluetooth hops



Non-Collaborative Coexistence Strategies

❑ Measure noise level and error rate:

Random bit errors $\neq$ Noise

1. Adaptive Packet Selection: Bluetooth uses coding (FEC and Modulation) depending upon interference. Use FEC only if noise. No FEC if interference.
2. Master Delay Policy: Bluetooth keeps track of error rates on various frequencies. Refrains from transmission on frequencies where interference is high
3. Adaptive frequency hopping: Hop over only good frequencies
4. Adaptive Notch Filter on WiFi

State

- ❑ Sensors periodically announce their “state”

30°F

55 km/hr

23 Kwh

- ❑ Small data transfers
- ❑ Event triggered or polled.
- ❑ Gateway devices, e.g., cell phones, can transfer this data to Internet.
- ❑ Sensors do not need to speak IP.

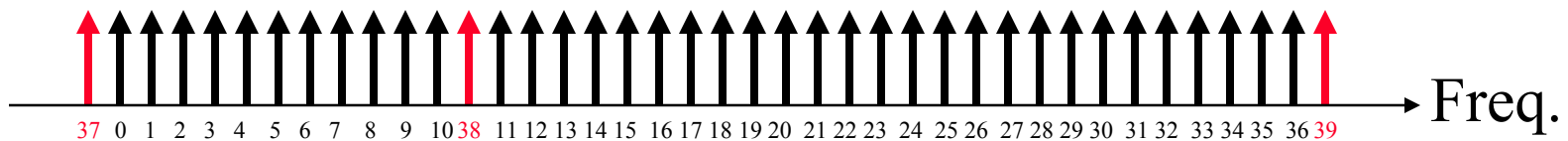
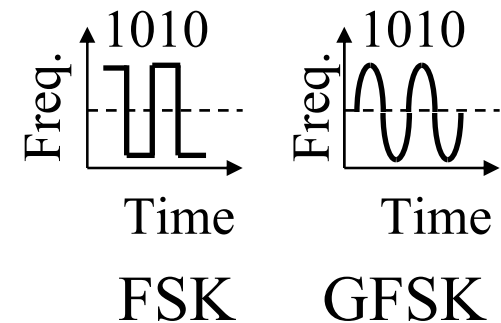


Bluetooth Smart

- ❑ **Low Energy:** 1% to 50% of Bluetooth classic
- ❑ **For short broadcast:** Your body temperature, Heart rate, Wearables, **sensors**, automotive, industrial.
Not for voice/video, file transfers, ...
- ❑ **Small messages:** 1Mbps data rate but throughput not critical.
- ❑ **Battery life:** In years from coin cells
- ❑ **Simple:** Star topology. No scatter nets, mesh, ...
- ❑ **Lower cost** than Bluetooth classic
- ❑ **New** protocol design based on Nokia's **WiBree** technology
Shares the same 2.4GHz radio as Bluetooth
⇒ Dual mode chips
- ❑ All new smart phones (iPhone, Android, ...) have dual-mode chips

Bluetooth Smart PHY

- ❑ 2.4 GHz. 150 m open field
- ❑ Star topology
- ❑ 1 Mbps Gaussian Frequency Shift Keying
Better range than Bluetooth classic
- ❑ Adaptive Frequency hopping. 40 Channels
with 2 MHz spacing.
- ❑ 3 channels reserved for **advertising** and 37 channels for data
- ❑ Advertising channels specially selected to avoid interference
with WiFi channels

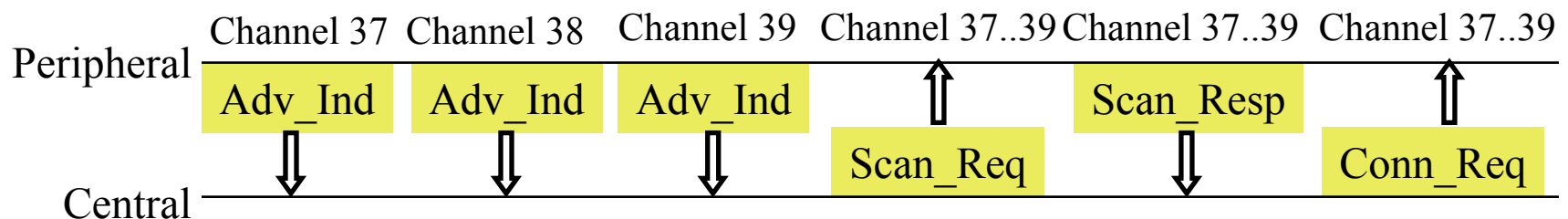


Ref: J. Decuir, "Bluetooth 4.0: Low Energy," 2010, <http://chapters.comsoc.org/vancouver/BTLER3.pdf>
 Washington University in St. Louis <http://www.cse.wustl.edu/~jain/cse574-16/>

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Bluetooth Smart MAC

- ❑ Two Device Types: “**Peripherals**” simpler than “**central**”
- ❑ Two PDU Types: Advertising, Data
- ❑ **Non-Connectable Advertising**: Broadcast data in clear
- ❑ **Discoverable Advertising**: Central may request more information. Peripheral can send data without connection
- ❑ **General Advertising**: Broadcast presense wanting to connect. Central may request a short connection.
- ❑ **Directed Advertising**: Transmit signed data to a previously connected master

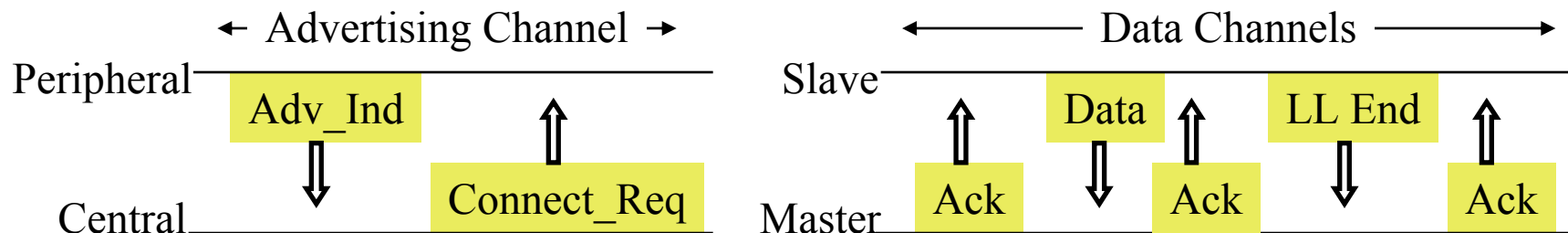


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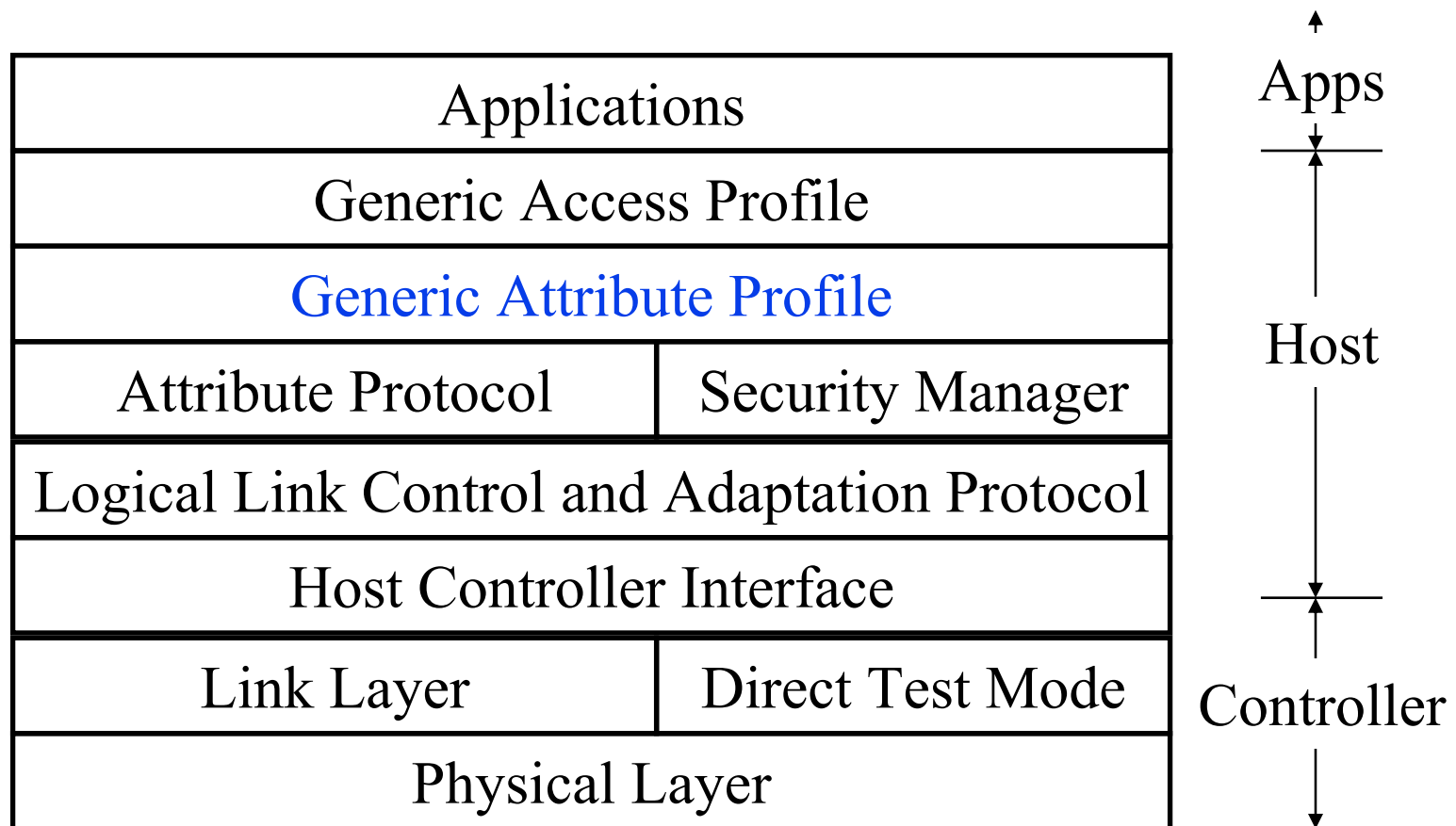
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Bluetooth Smart MAC (Cont)

- ❑ After connecting, master tells slave about hopping sequence and wake up cycle
- ❑ All subsequent data transfers in 37 data channels
- ❑ Both devices can sleep between transactions
- ❑ Data can be encrypted.
- ❑ ~3 ms per transaction, 15 mW Power = 10 mA using 1.5V
 - ↳ 30mAs/transaction
 - ↳ 21.6 M transactions using 180 mAh battery
 - ↳ 41.1 years with 1 transaction/minute



Bluetooth Smart Protocol Stack



Ref: J. Decuir, "Bluetooth 4.0: Low Energy," 2010, <http://chapters.comsoc.org/vancouver/BTLER3.pdf>
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Generic Attribute (GATT) Profile

- ❑ Defines data formats and interfaces with the Attribute Protocol
- ❑ Type-Length-Value (TLV) encoding is used
- ❑ Each attribute has a 16-bit Universally Unique ID (UUID) standardized by Bluetooth SIG
- ❑ 128-bit UUID if assigned by a manufacturer
- ❑ Allows any client to find a server, read/write data
Allows servers to talk to generic gateways
- ❑ Allows security up to AES-128
- ❑ Each to encode in XML
- ❑ Makes profile (application) development easier

Bluetooth Gateway Devices

- ❑ A gateway device helps connect a Bluetooth device to the Internet. Smart phone, Tablets, PC, ...
- ❑ A generic app can forward the data to the URL sent by the device



Bluetooth Smart Applications

- ❑ Proximity: In car, In room 303, In the mall
- ❑ Locator: Keys, watches, Animals
- ❑ Health devices: Heart rate monitor, physical activities monitors, thermometer
- ❑ Sensors: Temperature, Battery Status, tire pressure
- ❑ Remote control: Open/close locks, turn on lights

Ref: E. Vlugt, "Bluetooth Low Energy, Beacons and Retail," Verifone White paper, 2013, 12 pp.,
<http://www.verifone.com/media/3603729/bluetooth-low-energy-beacons-retail-wp.pdf>

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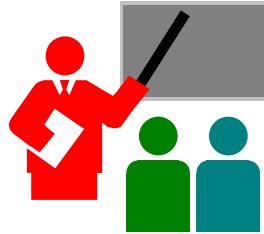


Beacons

- ❑ Advertizing based on proximity
- ❑ Peripherals (your phone) broadcasts its presence if Bluetooth is turned on
- ❑ Primary aim of these broadcasts is to allow device discovery
- ❑ Advertising packets consist of a header and max 27B of payload with multiple TLV-encoded data items
 - May include signal strength & Distance
- ❑ iOS7 iPhones can send/received iBeacons
- ❑ Can be used for customized advertising, indoor location, geofencing
- ❑ PayPal uses this to identify you.
You can pay using a PIN and your phone.



Summary



1. Bluetooth basic rate uses frequency hopping over 79 1-MHz channels with 1, 3, 5 slots packets.
2. Three inactive states: hold, sniff, park. Has a fixed set of applications called "Profiles"
3. Bluetooth and WiFi co-exist by time-sharing or adaptive frequency notching
4. Bluetooth Smart is designed for short broadcasts by sensors. 40 2-MHz channels with 3 channels reserved for advertising. One or two-message exchanges
5. Generic attribute profile allows new applications using UUID for data types

Homework 11

- ❑ Submit answer to the following Problem:
Assume that in one slot in Bluetooth 256 bits of payload could be transmitted. How many slots are needed if the payload size is (a) 512 bits, (b) 728 bits, and (c) 1024 bits. Assume that the non-payload portions do not change.

Reading List: Bluetooth

- ❑ Kevin Townsend, Carles Cufí, Akiba, Robert Davidson, "Getting Started with Bluetooth Low Energy," O'Reilly Media, Inc., May 2014, 180 pp., ISBN:978-1-4919-4951-1 (Safari Book), Chapter 2.
- ❑ J. Decuir, "Bluetooth 4.0: Low Energy," 2010, 62 pp., <http://chapters.comsoc.org/vancouver/BTLER3.pdf>
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Acronyms

❑ ACL	Asynchronous Connection List
❑ AES-128	Advanced Encryption Standard w 128 bit keys
❑ BLE	Bluetooth Low Energy
❑ BNEP	Bluetooth Network Encapsulation Protocol
❑ CAP	Connection Access Profile
❑ CSA	Core Specification Amendment
❑ dBm	Deci-bel milli-watt
❑ DPSK	Differential Phase Shift Keying
❑ EDR	Enhanced Data Rate,
❑ EU	European Union
❑ FEC	Forward Error Correction
❑ FSK	Frequency Shift Keying
❑ GATT	Generic Attribute
❑ GFSK	Gaussian Frequency Shift Keying
❑ GHz	Giga Hertz
❑ GPS	Global Positioning System
❑ HS	High Speed,

Acronyms (Cont)

❑ ID	Identifier
❑ IEEE	Institution of Electrical and Electronics Engineers
❑ IETF	Internet Engineering Task Force
❑ IG	Interest Group
❑ iOS	Apple's idevices Operating System
❑ IP	Internet Protocol
❑ IPv6	Internet Protocol version 6
❑ IrDA	Infrared Data Association
❑ IrMC	Infrared Mobile Communications
❑ IrOBEX	Infrared Object Exchange
❑ LAN	Local Area Network
❑ LAP	Lower address part
❑ LE	Low Energy
❑ LTE	Long Term Evolution
❑ MAC	Media Access Control
❑ MAN	Metropolitan Area Network

Acronyms (Cont)

❑ MB	Mega Byte
❑ MHz	Mega Hertz
❑ mW	milli Watt
❑ NAP	Non-significant address part
❑ OFDM	Orthogonal Frequency Division Multiplexing
❑ OUI	Organizationally Unique Identifier
❑ PAL	Protocol Adaptation Layer
❑ PAN	Personal Area Network
❑ PC	Personal Computer
❑ PDU	Protocol Data Unity
❑ PHY	Physical Layer
❑ PIN	Personal Identification Number
❑ RF	Radio Frequency
❑ RFCOMM	Radio Frequency Communication
❑ RFID	Radio Frequency Identifier
❑ SC	Standing Committee

Acronyms (Cont)

□ SCO	Synchronous Connection Oriented
□ SDP	Service Discovery Protocol
□ SG	Study Group
□ SIG	Special Interest Group
□ SIM	Subscriber Identity Module
□ TCS	Telephony Control Specification
□ TDD	Time-division duplex
□ TLV	Type-Length-Value
□ TV	Television
□ TX	Transmit
□ UAP	Upper address part
□ UCD	Unicast Connectionless Data
□ URL	Uniform Resource Locator
□ USB	Universal Serial Bus
□ UUID	Universally Unique Identifier
□ uW	Micro-Watt

Acronyms (Cont)

- ❑ WAN Wide Area Network
- ❑ WBS Wide Band Speed
- ❑ WiFi Wireless Fidelity
- ❑ WiMax Worldwide Interoperability for Microwave Access
- ❑ WPAN Wireless Personal Area Networks
- ❑ WRAN Wireless Regional Area Network
- ❑ XML Extensible Markup Language

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